Apparent Phantom Crossing as Template Bias: A Bounded Test Case with Λcos

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DESI DR2 reports a 2.8–4.2σ preference for dynamical dark energy in combined analyses, with standard $w_0 w_a$ fits favoring an apparent crossing of $w = -1$. We show that a non-phantom expansion history can produce apparent phantom crossings when analyzed with standard two-parameter templates. We construct a specific realization, Λcos, and constrain it using Pantheon+ and DESI DR2 BAO. At data-allowed parameter values, the induced CPL distortion is structurally present but modest ($w_0 \approx -1.02$) and of opposite sign in w_a, establishing the mechanism without reproducing the DESI best-fit amplitude.

The Λcos model yields $H^2(z)/H_0^2 = \alpha(1+z)^3 - \beta(1+z) + \Omega_\Lambda$, where α and β are determined by a single parameter s_0 and a fixed reference value $\Omega_\Lambda = 0.685$. Under the fiducial-matter diagnostic split, the model satisfies $w_{\rm eff}(z) > -1$ at all redshifts, recovers the fiducial flat ΛCDM limit as $s_0 \to 0$, and contains a negative $(1+z)^1$ term in H^2 absent from the four canonical FLRW density scalings. A joint MCMC fit to Pantheon+ (1701 SNe Ia) and DESI DR2 BAO (13 data points) yields $\Delta\chi^2 = +0.11$ relative to flat ΛCDM at the same parameter count, with $s_0 < 0.18$ (95% CL, flat prior). The constraint is robust across prior choices ($s_0 < 0.12$–0.21) and Ω_Λ variations (0.68–0.715). Adding compressed Planck distance priors shifts the preferred Ω_Λ in both ΛCDM and Λcos; allowing Ω_Λ to vary gives statistically equivalent fits ($\Delta\chi^2 = +0.28$ at equal parameter count).

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I. INTRODUCTION

The DESI collaboration's second data release [1] presents the most precise baryon acoustic oscillation measurements to date, spanning $0.3 < z < 2.3$ across seven tracer populations. Combined with Type Ia supernovae (Pantheon+ [2]) and CMB calibration (Planck [3]), the data favor a time-varying dark energy equation of state over the cosmological constant at 2.8–4.2σ within standard two-parameter parameterizations.

The inferred evolution follows the Chevallier-Polarski-Linder (CPL) parameterization w(a) = w_0 + w_a(1-a) [4, 5], with best-fit values suggesting $w_0 \approx$ -0.75 and $w_a \approx$ -0.86. This implies a crossing of w = -1 near z $\approx$ 0.5. In minimally coupled canonical single-field dark energy, crossing w = -1 is forbidden; realizing such behavior generally requires additional structure, such as noncanonical dynamics, multiple fields, interactions, or modified gravity [6, 7]. Cortês and Liddle [8] flag the crossing location at the center of the observable window as a "substantial and unsettling coincidence." The DESI extended analysis [9] reports consistent phantom-crossing behavior across five additional parameterizations (BA [10], JBP [11], exponential, logarithmic, and model-free binned).

The possibility that apparent phantom crossings can arise from parameterization choice has been recognized. Linder and Huterer [12] demonstrated that the CPL form introduces systematic bias when the true w(z) lies outside its functional family. Shafieloo, Sahni, and Starobinsky [13] showed that model-dependent reconstruction can produce spurious features, including phantom crossings, when applied to models outside the fitting basis. More recently, several groups have examined whether the DESI signal specifically could reflect such artifacts [8, 14].

The key question is not whether phantom crossings can be fitted to current data, but whether they are physically required, or instead arise as artifacts of the fitting template applied to a non-phantom expansion history. The present paper contributes to this discussion in two ways.

First, we construct a specific one-parameter expansion history, Λcos, whose diagnostic effective residual remains non-phantom under the fiducial-matter split, providing a concrete, falsifiable example rather than a generic cautionary argument. The model yields $\mathrm{H}^2(\mathrm{z})/\mathrm{H}_0^2$ = α(1+z)^3 - β(1+z) + Ω_Λ, for which $w_{\rm eff}$(z) > -1 follows analytically at all redshifts (Sec. III), and which recovers the fiducial flat ΛCDM limit as its single new parameter $s_0 \to 0$.

Second, we test the template bias mechanism quantitatively: we show that CPL, BA, and JBP all produce apparent phantom crossings when fitted to Λcos distances, and we scan the full parameter range to determine the amplitude of the induced distortion at data-allowed values. At the 95% upper limit $s_0 < 0.18$ from a joint Pantheon+ and DESI DR2 BAO fit, the CPL recovery gives $w_0 \approx$ -1.02 with positive w_a, establishing the structural mechanism while falling short of the DESI best-fit amplitude and sign.

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The goal is not to claim that Λcos explains the full DESI preference, but to provide a controlled test case in which the size and sign of template-induced phantom behavior can be computed and confronted with current data.

The paper is organized as follows. Section 2 constructs the Λcos model from a one-parameter background ansatz and selects the clock exponent by matter-era scaling. Section 3 proves $w_{\rm eff}$ > -1 at all redshifts. Section 4 demonstrates the template bias mechanism and scans its amplitude. Section 5 presents observational constraints including prior sensitivity, Ω_Λ robustness, and CMB distance priors. Section 6 collects predictions and falsification criteria. Section 7 discusses the results.

II. THE ΛCOS MODEL

A. Background ansatz

We construct a one-parameter deformation of the ΛCDM expansion history using a bounded auxiliary variable S ∈ (0, 1], related to redshift by 1 + z = s_0/S, where s_0 is the present-epoch value of S. Setting S = sin(t/2), where t is a phase variable advancing monotonically with cosmic time, satisfies three requirements: S is bounded above by unity, S → 0 at early times (t → 0), and the parameterization introduces exactly one new degree of freedom (s_0). We work on the monotonic branch 0 < t ≤ π, so that S = sin(t/2) increases from 0 to 1.

The redshift relation 1 + z = s_0/S reduces to the standard FLRW form a/a_0 = S/s_0 with the identification a ∝ S. The ΛCDM limit corresponds to s_0 → 0 with S/s_0 held fixed, in which the trigonometric corrections vanish and S/s_0 reduces to a conventional scale factor.

The choice of sin(t/2) over other bounded functions (e.g., tanh, logistic, or polynomial) is motivated by algebraic simplicity: the resulting H^2(z) is a finite polynomial in (1+z) with coefficients determined by s_0 alone (Sec. II C). Alternative bounded parameterizations would generally produce transcendental or infinite-series expressions for H^2(z), complicating both the analytic proof of $w_{\rm eff}$ > -1 (Sec. III) and the comparison with standard FLRW components.

B. Clock selection

The phase variable t and proper time τ are related by a clock rate. From S = sin(t/2), the non-vacuum scaling of the expansion rate is governed by:

$$\frac{1}{S}\frac{dS}{d\tau} = \frac{\cos(t/2)}{2\sin(t/2)} \cdot \frac{dt}{d\tau}$$

This quantity is not yet the physical Hubble parameter; it defines the redshift scaling assigned to the non-vacuum sector before the cosmological constant is added (Sec. II C).

For a power-law clock dt/dτ = S^n, the non-vacuum scaling gives (1/S)(dS/dτ) = 1/2 cos(t/2) S^{n-1}. The power-law form is the minimal one-parameter family relating phase and proper time; more general clocks dt/dτ = f(S) would introduce functional freedom beyond a single exponent and are not considered here. We adopt the standard Friedmann scaling $H^2 \propto \rho_{\rm m} \propto S^{-3}$ [15] as a selection criterion, requiring consistency with the observed matter-dominated expansion at high redshift. At high redshift (S → 0), cos(t/2) → 1, and the requirement (1/S)(dS/dτ) ∝ $S^{-3/2}$ selects n - 1 = -3/2, giving:

$$\frac{dt}{d\tau} = S^{-1/2} = \sin^{-1/2}(t/2)$$

With this clock, the exact non-vacuum kernel is:

$$\frac{1}{S}\frac{dS}{d\tau} = \frac{\sqrt{1-S^2}}{2S^{3/2}}$$

The $\sqrt{(1-S^2)}$ factor equals unity at high z and generates the $(1+z)^1$ correction at low z (Sec. II D). For the tested integer alternatives, n = +1, 0, -1 give (1/S)(dS/dτ) ∝ $(1+z)^0$, $(1+z)^1$, $(1+z)^2$, respectively. Only n = -1/2 gives the matter-era scaling $(1+z)^3/^2$. This was verified against Pantheon+ data, where all three integer-power alternatives give $\Delta\chi^2$ > 60 relative to ΛCDM (Appendix A).

C. The Hubble rate

Using 1 + z = s_0/S, we write S = s_0/(1+z). Squaring the non-vacuum kernel from Sec. II B and substituting:

$$\mathcal{K}(z;\, s_0) = \frac{(1+z)^3 - s_0^2(1+z)}{1-s_0^2}$$

where the normalization $\mathcal{K}(0;\, s_0) = 1$ is enforced by the denominator. The cosmological constant, as a vacuum energy density independent of the expansion history, enters the Friedmann equation [15] as a separate additive component. The physical dimensionless Hubble rate is then:

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = (1-\Omega_\Lambda)\,\mathcal{K}(z;\, s_0) + \Omega_\Lambda$$

Ω_Λ = 0.685 is taken as a fixed reference value; Sec. V D demonstrates stability of all results across the range 0.68–0.715. Expanding:

$$\frac{H^2(z)}{H_0^2} = \frac{1-\Omega_\Lambda}{1-s_0^2}(1+z)^3 - \frac{(1-\Omega_\Lambda)\,s_0^2}{1-s_0^2}(1+z) + \Omega_\Lambda$$

Defining $\alpha \equiv (1\text{-}\Omega_\Lambda)/(1\text{-s}_0^2)$ and $\beta \equiv (1\text{-}\Omega_\Lambda)\text{s}_0^2/(1\text{-s}_0^2)$, the expression reads $\text{H}^2/\text{H}_0^2 = \alpha(1\text{+z})^3$ - $\beta(1\text{+z}) + \Omega_\Lambda$, with α - $\beta + \Omega_\Lambda = 1$ enforcing E(0) = 1. Radiation ($\Omega_r \approx 9 \times 10^{-5}$) is omitted; its contribution is below 0.3% at all redshifts probed by the datasets used here (z < 2.4).

In the baseline fits, Ω_Λ is fixed to the reference value above, so the cosmological degree of freedom replacing Ω_m is s_0. The baseline Λcos and flat ΛCDM comparisons therefore use the same number of fitted parameters: s_0, $H_0 r_d$, and M_B for Λcos; Ω_m, $H_0 r_d$, and M_B for flat ΛCDM. The fiducial flat ΛCDM expansion history is recovered as $\text{s}_0 \to 0$, where $\alpha \to 1$ - Ω_Λ and $\beta \to 0$.

D. The $(1+z)^1$ term

The negative $(1\text{+z})^1$ term is absent from the four canonical FLRW components:

TABLE I. FLRW density components compared with the bounded $(1+z)^1$ correction term in Λcos.

Component	Redshift scaling
Radiation	$(1\text{+z})^4$
Matter	$(1\text{+z})^3$
Curvature	$(1\text{+z})^2$
Cosmological constant	$(1\text{+z})^0$
Λcos correction	**$(1\text{+z})^1$**

A constant-w fluid with w = -2/3 (domain walls) would produce the same redshift scaling. The Λcos term is distinguished here by the tied coefficient -$\beta(\text{s}_0)$, rather than by the scaling alone. Its coefficient is -β = -(1-$\Omega_\Lambda)\text{s}_0^2/(1\text{-s}_0^2)$, strictly negative for $\text{s}_0 > 0$ and vanishing in the ΛCDM limit. Interpreted as an effective fluid, the negative coefficient would imply a violation of the weak energy condition. However, the $(1\text{+z})^1$ term is not an independent component but a correction arising from the bounded parameterization of the matter sector; energy condition arguments apply to physical stress-energy, not to diagnostic residuals of an algebraic decomposition.

III. w_{eff}(Z) > -1 AT ALL REDSHIFTS

To define a diagnostic effective dark energy equation of state, one must specify a matter subtraction. We adopt the split in which deviations from the fiducial matter scaling are assigned to the effective dark-energy sector, using $\Omega_m = 1$ - $\Omega_\Lambda = 0.315$ (the fiducial matter fraction, independent of s_0) rather than the dressed coefficient $\alpha = \Omega_m/(1\text{-s}_0^2)$ that appears in the $(1\text{+z})^3$ term.

The effective dark energy density is then:

$$\begin{aligned} X(z) &= E^2(z) - \Omega_m(1+z)^3 \\ &= \frac{\Omega_m\, s_0^2}{1-s_0^2}\left[(1+z)^3 - (1+z)\right] + \Omega_\Lambda \end{aligned}$$

Assuming this effective component satisfies the standard continuity equation $\text{d}\rho_{\text{DE}}/\text{dz} = 3(1\text{+}w_{\text{eff}})\rho_{\text{DE}}/(1\text{+z})$, the diagnostic equation of state is:

$$w_{\text{eff}}(z) = -1 + \frac{(1+z)}{3X}\frac{dX}{dz}$$

Computing the derivative:

$$\frac{dX}{dz} = \frac{\Omega_m\, s_0^2}{1-s_0^2}\left[3(1+z)^2 - 1\right]$$

The correction to w = -1 is therefore:

$$w_{\text{eff}}(z)+1 = \frac{\Omega_m\, s_0^2\,(1+z)\left[3(1+z)^2-1\right]}{3(1-s_0^2)\left[\frac{\Omega_m\, s_0^2}{1-s_0^2}\left((1+z)^3-(1+z)\right)+\Omega_\Lambda\right]}$$

For z $\geq$ 0 (corresponding to S $\leq \text{s}_0$) and $\text{s}_0 \in (0, 1)$:

- **Numerator:** $3(1\text{+z})^2$ - 1 $\geq$ 2 (equality at z = 0). Strictly positive.
- **Denominator:** $(1\text{+z})^3$ - (1+z) = (1+z)(z)(2+z) $\geq$ 0 for z $\geq$ 0, and the additive $\Omega_\Lambda > 0$ ensures X(z) is strictly positive at every z.

Therefore:

$$\boxed{w_{\text{eff}}(z) > -1}$$

for the fiducial-matter diagnostic split, for all $z \geq 0$, $s_0 \in (0, 1)$.

The constraint $0 < \text{s}_0 < 1$ follows from the definition $\text{s}_0 = \sin(t_{\text{now}}/2)$ with $0 < t_{\text{now}} \leq \pi$ (Sec. II A). At s_0 = 0, w_{eff} = -1 everywhere (the fiducial flat ΛCDM limit). For $\text{s}_0 > 0$, the correction is positive and begins at order s_0^2.

The decomposition choice matters: had we instead subtracted the dressed coefficient $\alpha(1\text{+z})^3$, the residual would be -$\beta(1\text{+z}) + \Omega_\Lambda$, which is a decreasing function of z and yields a different w_{eff} trajectory. Because that residual dilutes faster than a cosmological constant, it necessarily produces w_{eff} < -1 at sufficiently high z; this is a decomposition artifact, not a property of the physics. The motivation for the adopted diagnostic split is that $\Omega_m = 1$ - Ω_Λ is held fixed as the fiducial matter fraction, independent of s_0. The s_0-dependent excess relative to $\Omega_m(1\text{+z})^3$ is therefore assigned to X(z), the effective residual component whose equation of state is being diagnosed.

IV. TEMPLATE BIAS DEMONSTRATION

A. Method

To test whether standard w(z) parameterizations can produce apparent phantom crossings from Λcos distances, we generate noise-free BAO observables (D_M/r_d, D_H/r_d, D_V/r_d) from the Λcos model at the seven DESI DR2 effective redshifts (z = 0.295, 0.510, 0.706, 0.934, 1.321, 1.484, 2.330), using H_0 = 67.4 km/s/Mpc and r_d = 147.1 Mpc. We then fit four w(z) parameterizations to these mock observables using the published DESI covariance structure, with $\Omega_m = 0.315$ held fixed to match the input Λcos model, isolating the effect of the w(z) parameterization. For each effective redshift we use the same observable set and covariance block structure as the DESI DR2 compressed BAO likelihood; when multiple observables are reported at the same redshift, their published correlation coefficients are included.

The four parameterizations are CPL [4, 5], Barboza-Alcaniz (BA) [10], Jassal-Bagla-Padmanabhan (JBP) [11], and a three-parameter polynomial in (1-a). Each is a standard fitting form used in the DESI extended analysis [9].

The mock exercise uses $s_0 = 0.389$ to maximize visibility of the effect. The template bias mechanism operates at any $s_0 > 0$ (see Sec. IV C). The fits minimize $\chi^2 = \Delta^T C^{-1}\Delta$, where Δ is the difference between model and mock BAO observables and C is the DESI DR2 covariance matrix constructed from the published uncertainties and correlations. The χ^2 values reported below reflect fits to noise-free mock data with DESI covariance weighting.

B. Results

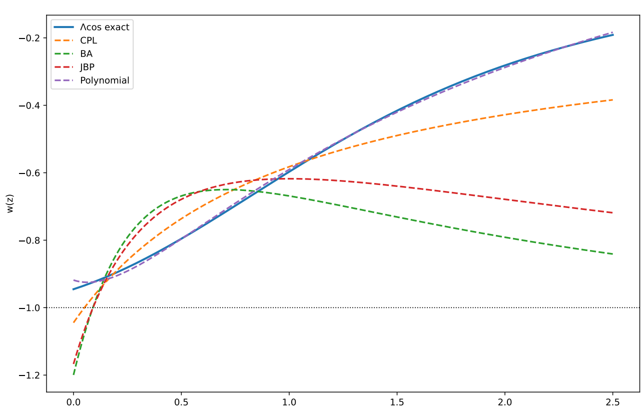

FIG. 1. Exact Λcos $w_{\rm eff}$(z) (solid, always > -1) overlaid with recovered w(z) from all four parameterizations. CPL, BA, and JBP dip below w = -1 at low z. The three-parameter polynomial tracks the true curve without crossing.

Three of four parameterizations produce phantom crossings from input that satisfies w > -1 everywhere (Sec. III). The polynomial, with one additional free parameter, does not produce a phantom crossing in this case. The effect arises from projecting the curvature of the Λcos distance-redshift relation onto a restricted two-parameter basis, absorbing the residual into phantom-crossing parameter values.

C. Threshold scan

To determine the amplitude of the template bias at data-allowed values of s_0, we repeat the CPL fit across $s_0 \in [0.01, 0.40]$ in steps of 0.01. The CPL recovery produces an apparent phantom-crossing trajectory at every s_0 tested, including the smallest value $s_0 = 0.01$: the fitted $w_0 <$ -1 at low redshift, with the positive w_a driving w back above -1 at higher redshift. Representative results:

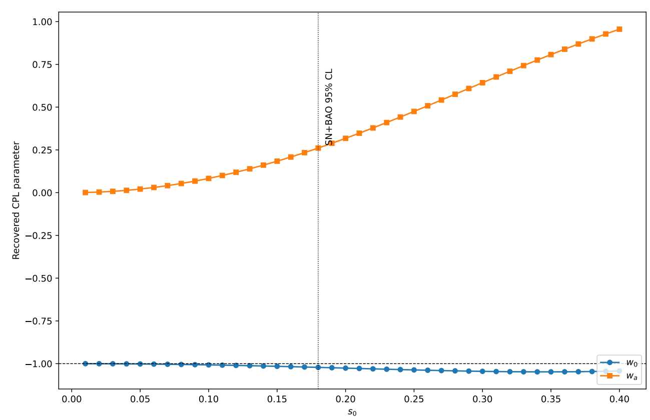

FIG. 2. Recovered CPL parameters w_0 (circles) and w_a (squares) as a function of s_0. The apparent crossing trajectory persists at every tested $s_0 > 0$. The horizontal dashed line marks w = -1.

Two features are evident. First, the crossing persists at every tested $s_0 > 0$, confirming that the template bias is a structural property of the CPL basis applied to non-phantom models of this type, not an artifact of a particular parameter value. Second, at the SN+BAO 95% upper limit $s_0 < 0.18$ (Sec. V), the induced distortion is modest: $w_0 \approx$ -1.02 with $w_a \approx$ +0.26.

D. Comparison with the DESI best fit

The DESI CPL best fit reports $w_0 \approx$ -0.75 and $w_a \approx$ -0.86. The Λcos-induced distortion at data-allowed s_0 differs in both amplitude ($|w_0 + 1|$ = 0.02 vs 0.25, a factor of ~12) and sign (w_a = +0.26 vs -0.86).

TABLE II. Best-fit $w(z)$ parameters from CPL, BA, JBP, and a three-parameter polynomial fitted to noise-free Λcos mock BAO at $s_0 = 0.389$.

Parameterization	Best-fit parameters	χ^2	Crosses w = -1?
CPL: $w_0 + w_a$(1-a)	w_0 = -1.056, w_a = 0.951	1.0	Yes
BA: $w_0 + w_a$ a(1-a)/(a^2-2a+2)	w_0 = -1.295, w_a = 3.315	10.8	Yes
JBP: $w_0 + w_a$ a(1-a)	w_0 = -1.226, w_a = 2.551	6.1	Yes
Polynomial: $w_0 + w_1$(1-a) + w_2(1-a)^2	w_0 = -0.927, w_1 = -0.162, w_2 = 1.637	0.2	No

TABLE III. Recovered CPL parameters as a function of s_0, illustrating that the apparent crossing trajectory persists at every tested $s_0 > 0$.

s_0	w_0	w_a	w_0 deviation from -1
0.01	-1.00007	+0.001	7×10^{-5}
0.06	-1.002	+0.021	0.002
0.18	-1.022	+0.261	0.022
0.39	-1.044	+0.928	0.044
DESI observed	-0.75	-0.86	0.25

The template bias mechanism is therefore established as a structural effect, but the Λcos model at its current constraint does not reproduce the DESI best-fit parameters quantitatively. Bridging the amplitude gap would require either a larger bounded deformation ($s_0 \gtrsim 0.4$, ruled out by current data) or a different functional form for the effective dark-energy sector.

This distinction is important. The paper demonstrates that a non-phantom expansion history (Λcos) produces apparent phantom crossings under standard two-parameter templates. The claim is not that Λcos at $s_0 < 0.18$ explains the specific amplitude of the DESI signal.

V. OBSERVATIONAL CONSTRAINTS

A. Data

We fit jointly to the Pantheon+ supernova compilation (1701 SNe Ia with full statistical and systematic covariance [2]) and DESI DR2 BAO measurements (1 isotropic D_V/r_d at z = 0.295 and 6 anisotropic D_M/r_d, D_H/r_d pairs at z = 0.51–2.33, with published cross-correlations [1]). The total dataset comprises 1714 observables (1701 supernova magnitudes and 13 BAO measurements).

The supernova likelihood is $\chi^2_{\rm SN} = \Delta^T \mathrm{C}^{-1}\Delta$, where Δ_i = m_{b,i} - M_B - $\mu(z_i)$ and C is the 1701 × 1701 covariance matrix. The BAO likelihood is constructed from the published uncertainties and inter-observable correlations at each redshift bin, with bins treated as independent. The total likelihood is $\chi^2 = \chi^2_{\rm SN} + \chi^2_{\rm BAO}$.

B. Primary fit: SN+BAO

MCMC sampling uses an affine-invariant ensemble sampler (32 walkers, 5000 steps, 1000 burn-in). Convergence is confirmed via integrated autocorrelation time ($\tau_{\rm max} \approx 46$ for Λcos and $\tau_{\rm max} \approx 36$ for ΛCDM, with the post-burn chain length of $4000 \gtrsim 85\tau$ for all parameters).

Flat ΛCDM free parameters: Ω_m, $H_0 r_d$, M_B. Λcos free parameters: s_0, $H_0 r_d$, M_B (with $\Omega_\Lambda = 0.685$ fixed). Both models have three free parameters. The primary results are reported at the fixed reference value $\Omega_\Lambda = 0.685$; Sec. V D demonstrates stability of the s_0 constraint across the range 0.68–0.715, and Sec. V E confirms the model remains competitive when Ω_Λ is freed entirely ($\Delta\chi^2 = +0.28$ at equal parameter count).

Uncertainties are posterior median and 68% credible intervals. The Λcos best-fit sits at $s_0 = 0.001$ (the numerical lower bound of the prior; the analytic ΛCDM limit is $s_0 = 0$). The posterior median of 0.076 reflects prior volume at $s_0 > 0$. The 95th percentile gives:

$$s_0 < 0.18 \quad \text{(95\% CL, flat prior).}$$

The $\Delta\chi^2 = +0.11$ at equal parameter count corresponds to ΔAIC = ΔBIC = +0.11: no preference for either model.

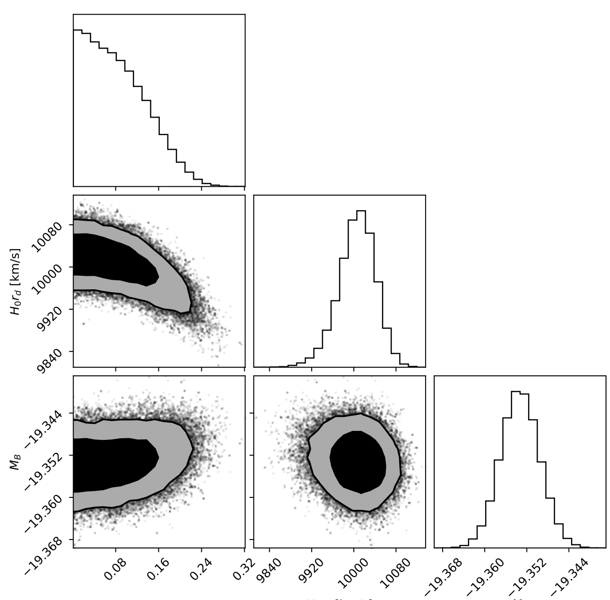

FIG. 3. Corner plot of the Λcos posterior in (s_0, $H_0 r_d$, M_B) space from the SN+BAO fit.

TABLE IV. Joint Pantheon+ + DESI DR2 BAO posterior summaries for flat ΛCDM and Λcos.

	Flat ΛCDM	Λcos
Primary parameter	$\Omega_m = 0.312$ [0.304, 0.321]	$s_0 = 0.076$ [0.023, 0.143]
$H_0 r_d$ **(km/s)**	10043 [9977, 10111]	10008 [9972, 10040]
M_B	-19.355 [-19.360, -19.350]	-19.353 [-19.357, -19.350]
$\chi^2_{\rm min}$	1772.5	1772.6
$\chi^2_{\rm SN}$ / $\chi^2_{\rm BAO}$	1759.9 / 12.6	1759.0 / 13.5

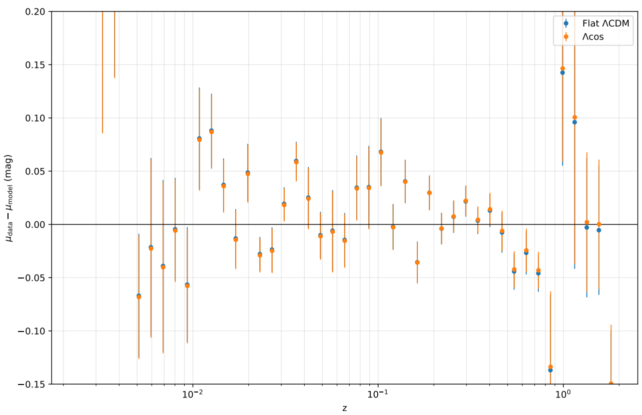

FIG. 4. Pantheon+ binned residuals ($\mu_{\rm data}$ - $\mu_{\rm model}$) for ΛCDM and Λcos at joint best-fit parameters. The two models are indistinguishable.

C. Prior sensitivity

Since the Λcos posterior is concentrated near the lower prior boundary, the 95% upper limit is prior-sensitive. We test two alternative priors by reweighting the baseline chain:

The $\chi^2_{\rm min}$ is identical across all three priors (the likelihood is unchanged; only the posterior weighting differs). The 95% upper limit ranges from 0.12 to 0.21, a factor of ˜1.8. The constraint is prior-sensitive in detail but data-driven in character: all three priors yield $s_0 \ll 1$.

D. Ω_Λ sensitivity

The primary fit uses $\Omega_\Lambda = 0.685$ as a fixed reference value. To test robustness, we repeat the SN+BAO fit at four alternative values spanning the range between the SN+BAO-preferred and CMB-preferred regions:

The s_0 constraint varies smoothly with Ω_Λ, with the 95% upper limit ranging from 0.16 to 0.35 across the tested interval. The best $\Delta\chi^2$ occurs near $\Omega_\Lambda = 0.69$. At the CMB-preferred value $\Omega_\Lambda = 0.715$, the best-fit s_0 shifts to 0.288 (nonzero) and the model is mildly disfavored ($\Delta\chi^2 = +2.38$). The results are stable across the tested range; no fine-tuning of Ω_Λ is required.

E. CMB distance priors

The DESI DR2 phantom-crossing result was obtained jointly with Planck CMB data. To test whether Λcos is compatible with CMB constraints, we add compressed Planck 2018 distance priors (shift parameter R = 1.7502 ± 0.0046, acoustic scale $\ell_A = 301.47 \pm 0.09$, treated as independent Gaussians) to the SN+BAO likelihood. These quantities constrain the integral $\int$dz/E(z) to the last-scattering surface at z* = 1090. Radiation (Ω_r = 9.15 × 10^{-5}) is included for this integral. Non-flat ΛCDM includes a curvature term Ω_k = 1 - Ω_m - Ω_Λ. Since the Planck distance priors are derived within a restricted background model space, their use here should be understood as a consistency diagnostic rather than a full CMB likelihood evaluation.

The four-parameter Λcos fit with Ω_Λ free and the four-parameter non-flat ΛCDM fit provide the primary comparison:

The two models are indistinguishable at $\Delta\chi^2 = +0.28$ with the same parameter count.

For comparison, fixing $\Omega_\Lambda = 0.685$ introduces a penalty of $\Delta\chi^2 \approx +66$ relative to flat ΛCDM (3 parameters each). This tension is driven by the BAO sector ($\chi^2_{\rm BAO}$ = 120 vs 30), not by the CMB priors themselves ($\chi^2_{\rm CMB}$ = 1.3 for Λcos vs 14.7 for ΛCDM). The effect is not specific to Λcos: adding CMB priors shifts the preferred Ω_Λ from 0.685–0.688 (SN+BAO) to ≈ 0.714 (Λcos with Ω_Λ free) and ≈ 0.720 (non-flat ΛCDM). Flat ΛCDM absorbs the shift through Ω_m (which moves from 0.312 to 0.286). Λcos at fixed Ω_Λ cannot absorb it. The shift from $\Omega_\Lambda \approx$ 0.685 to $\Omega_\Lambda \approx 0.714$ should therefore be interpreted as a background-distance consistency issue in the compressed-prior setup, rather than as a failure mode unique to Λcos.

The compressed-prior test used here is an approximate background-level consistency check, not a substitute for the full correlated Planck likelihood.

F. Parameter accounting

The matter fraction in Λcos is set through Ω_m = 1 - Ω_Λ at the chosen reference value. The $(1+z)^3$ coefficient $\alpha = \Omega_m/(1-s_0^2)$ is a dressed version of this input. As a consistency check: at any s_0, $\alpha(1 - s_0^2)$ returns Ω_m = 0.315. The dressing is algebraically self-consistent and does not introduce additional freedom.

TABLE V. Prior sensitivity of the Λcos s_0 constraint across three prior choices.

Prior on s_0	s_0 median	s_0 95% UL	$\chi^2_{\rm min}$
Flat $s_0 \in [0.001, 0.99]$ (baseline)	0.076	0.185	1772.6
Flat in s_0^2 (more weight at larger s_0)	0.115	0.208	1772.6
Flat in $\log_{10}(s_0)$ (scale-invariant)	0.011	0.115	1772.6

TABLE VI. Λcos sensitivity to the Ω_Λ reference value across the SN+BAO and CMB-preferred range.

Ω_Λ	s_0 median	s_0 95% UL	$\Delta\chi^2$ vs ΛCDM
0.680	0.063	0.165	+0.83
0.685 (baseline)	0.076	0.185	+0.11
0.690	0.089	0.205	+0.07
0.700	0.147	0.260	+0.96
0.715 (CMB-preferred)	0.278	0.349	+2.38

TABLE VII. Comparison of Λcos with Ω_Λ free against non-flat ΛCDM after adding compressed Planck distance priors.

	Λcos (Ω_Λ free)	Non-flat ΛCDM
Free params	s_0, Ω_Λ, $H_0 r_d$, M_B	Ω_m, Ω_Λ, $H_0 r_d$, M_B
Best-fit Ω_Λ	0.714 [0.711, 0.718]	0.720 [0.708, 0.732]
$\chi^2_{\rm min}$	1814.8	1814.5
$\Delta\chi^2$	+0.28	0 (reference)

G. Model comparison

To contextualize the Λcos result, we compare against wCDM (constant w as a free parameter, 4 free parameters: Ω_m, w, $H_0 r_d$, M_B) fitted to the same SN+BAO dataset:

TABLE VIII. Model comparison against Pantheon+ + DESI DR2 BAO: flat ΛCDM, Λcos at two Ω_Λ values, and wCDM.

Model	Free params	$\chi^2_{\rm min}$	$\Delta\chi^2$	ΔAIC	ΔBIC
Flat ΛCDM	3	1772.4	0	0	0
Λcos ($\Omega_\Lambda = 0.685$)	3	1772.6	+0.11	+0.11	+0.11
Λcos ($\Omega_\Lambda = 0.715$)	3	1774.8	+2.38	+2.38	+2.38
wCDM	4	1759.4	-13.0	-11.0	-5.6

The wCDM fit yields w = -0.855 [-0.89, -0.82] (68% CI), preferring a quintessence-like (w > -1) departure from ΛCDM at ΔAIC = -11.0. This is the same qualitative signal underlying the DESI phantom-crossing result: the SN+BAO data favor some departure from w = -1. The distinction is that wCDM captures this as a constant shift toward w > -1, while CPL projects it into a crossing through w = -1. The Λcos model, for which $w_{\rm eff}$ > -1 at all redshifts (Sec. III), is consistent with the wCDM direction but does not achieve the same χ^2 improvement because the data-allowed s_0 is too small to produce a detectable departure.

For the baseline flat prior in s_0, a Savage-Dickey density ratio at the s_0 prior boundary (with boundary-reflected KDE) gives $B_{01} \approx 7.1$ (stable to ±0.02 across bandwidths), corresponding to moderate evidence for ΛCDM over Λcos on the Jeffreys scale. This Bayes factor should be interpreted conditional on the baseline prior choice. The result confirms the $\Delta\chi^2$ finding: current data do not require the ansatz's correction term.

VI. PREDICTIONS AND FALSIFICATION CRITERIA

A. The $(1+z)^1$ signature

The joint SN+BAO fit constrains $|\beta| < 0.011$ at 95% CL (from $s_0 < 0.18$ with $\Omega_m = 0.315$). The predicted fractional contribution to $H^2(z)$ at z = 1 is:

$$\frac{|\beta|(1+z)}{E^2(z)} < 0.7\% \quad (95\%\ \text{CL at}\ z = 1).$$

This is below the precision of current BAO measurements (DESI DR2: ˜2–3% per bin in D_H/r_d) but potentially within reach of the next generation. Euclid DR1 spectroscopic BAO (z = 0.9–1.8, four redshift bins) is forecast at 1–2% per bin [16], marginal for detection. The discriminating regime is DESI full-survey and Euclid DR2 at ˜0.5% per bin, where the $(1+z)^1$ signature reaches approximately 1.5σ per bin for s_0 ˜ 0.15, with correlated bins improving aggregate sensitivity. A definitive test of the full $s_0 > 0.10$ range requires next-generation spectroscopic surveys (MegaMapper, MUST, WST) forecast at sub-0.5% per-bin precision [17].

A statistically significant detection of a negative $(1+z)^1$ component in $H^2(z)$ would constitute evidence for a non-standard contribution to the expansion history. While a domain-wall fluid (w = -2/3) produces the same scaling, the Λcos term is distinguished by the tied coefficient $-\beta(s_0)$ rather than by an independent energy density. Previous template-bias analyses identify the mechanism generically; the present model commits to a specific functional form whose coefficient is determined by a single parameter already constrained by current data.

TABLE IX. Falsification criteria for the three load-bearing Λcos predictions.

Observable	Λcos prediction	Falsified if
$(1+z)^1$ coefficient in H^2	Non-positive, magnitude tied to s_0	Positive coefficient detected, or negative coefficient inconsistent with fitted s_0
Diagnostic $w_{\text{eff}}(z)$	$>$ -1 (fiducial split, any $s_0 > 0$)	Same split gives $w_{\text{eff}} <$ -1 at $\geq 2\sigma$
ΛCDM limit	$s_0 = 0$ recovers fiducial flat ΛCDM	(none required)

B. Falsification criteria

VII. DISCUSSION

A. Template bias

The central result of this paper is that a non-phantom expansion history can produce apparent phantom crossings when fitted with standard two-parameter w(z) templates. We have demonstrated this explicitly for CPL, BA, and JBP parameterizations applied to Λcos distances: all three recover $w_0 <$ -1 from Λcos distances for which $w_{\text{eff}} >$ -1 at every redshift under the fiducial-matter diagnostic split (Sec. III). A three-parameter polynomial does not produce the crossing, confirming that the effect arises from basis restriction rather than from the underlying expansion history.

This finding has precedent. Linder and Huterer [12] identified CPL template bias, Shafieloo, Sahni, and Starobinsky [13] demonstrated spurious phantom crossings from model-dependent reconstruction, and recent work [8, 14] has raised similar concerns about the DESI signal specifically. The present contribution is to provide a concrete one-parameter non-phantom model (rather than a generic cautionary argument), to test multiple parameterizations, and to scan the template bias amplitude across the full parameter range allowed by current data.

The DESI extended analysis [9] itself tests five parameterizations beyond CPL and finds consistent phantom-crossing behavior across all of them. This might seem to rule out template bias as an explanation. However, the DESI analysis fits each parameterization to real data, which may contain a genuine departure from ΛCDM (as the wCDM comparison in Sec. V G suggests). The mock exercise in Sec. IV asks a different question: can a model that provably satisfies w $>$ -1 produce phantom crossings across the same family of templates? The answer is yes, confirming that template agreement across parameterizations is necessary but not sufficient evidence for a true crossing. The convention-dependence of the phantom classification is itself part of the finding: the same expansion history is diagnosed as phantom or non-phantom depending on the matter subtraction, which is precisely the ambiguity that template bias exploits.

At data-allowed values ($s_0 < 0.18$), the induced CPL distortion is $w_0 \approx$ -1.02 with positive w_a, structurally present but modest in amplitude and opposite in sign to the DESI best fit ($w_0 \approx$ -0.75, $w_a \approx$ -0.86). The paper establishes the mechanism without claiming to reproduce the DESI signal quantitatively.

B. Status of Λcos

The Λcos model rests on three inputs: a bounded auxiliary variable $S = \sin(t/2)$ with redshift relation $1 + z = s_0/S$ (Sec. II A), the power-law clock rate $dt/d\tau = S^{-1}/^2$ selected by matter-era asymptotic scaling (Sec. II B), and a fixed reference value $\Omega_\Lambda = 0.685$ (Sec. II C). The resulting $H^2(z)$ is derived algebraically from these inputs. The fiducial flat ΛCDM expansion history is recovered as $s_0 \to 0$.

The present analysis constrains only background observables (distances and expansion rates). The Λcos modification to H(z) also affects the linear growth rate $f(z)\sigma_8(z)$, which provides an independent test. However, predicting growth requires specifying the perturbation-level sound speed of the effective dark energy component, which the background ansatz does not constrain. A comparison with redshift-space distortion data (e.g., $f\sigma_8$ from DESI or earlier surveys) is deferred to future work.

Current combined data (Pantheon+ and DESI DR2 BAO) constrain $s_0 < 0.18$ at 95% CL, consistent with the ΛCDM limit. Λcos matches ΛCDM at $\Delta\chi^2 = +0.11$ across the SN+BAO dataset. Adding compressed Planck distance priors shifts the preferred Ω_Λ, a background-distance consistency issue that is removed when Ω_Λ is freed ($\Delta\chi^2 = +0.28$ at equal parameter count). The parity with ΛCDM at current precision is a feature of the model's design: the fiducial ΛCDM limit is recovered exactly as $s_0 \to 0$, so agreement with existing data is expected. The $(1+z)^1$ term becomes detectable only at the ˜1% precision level in $H^2(z)$, which is where Λcos and ΛCDM diverge. The model is a minimal one-parameter deformation of the fiducial flat ΛCDM expansion history, with a specific falsifiable signature (the negative $(1+z)^1$ term) testable by next-generation BAO measurements.

VIII. CONCLUSIONS

A non-phantom expansion history can produce apparent phantom crossings under standard two-parameter templates. The Λcos model, constructed as a one-parameter deformation of the fiducial flat ΛCDM expansion history using a bounded auxiliary variable, provides a concrete realization for which $w_{\rm eff}$(z) > -1 at all redshifts under the fiducial-matter diagnostic split, while matching the Pantheon+ and DESI DR2 BAO distance-redshift relation at $\Delta\chi^2 = +0.11$ relative to flat ΛCDM. Current data constrain the model's single new parameter to $s_0 < 0.18$ at 95% confidence, and the predicted negative $(1+z)^1$ contribution to $H^2(z)$ is a target for next-generation BAO measurements.

IX. DATA AND CODE AVAILABILITY

The MCMC chains, analysis scripts, and figure-generation code that reproduce all results in this paper are publicly available at github.com/dmobius3/lambda-cos and archived at Zenodo (DOI:10.5281/zenodo.19798852). The Pantheon+ supernova compilation [2] is available at pantheonplussh0es.github.io and the DESI DR2 BAO measurements [1] at data.desi.lbl.gov.

X. APPENDIX A: CLOCK EXPONENT SELECTION

Three alternative clock rates were tested against the joint Pantheon+ + DESI DR2 BAO dataset using the same MCMC setup as the primary Λcos fit (Sec. V B): identical priors on $H_0 r_d$ and M_B, identical sampler configuration, identical likelihood construction.

*Model B saturates at the s_0 prior floor. The likelihood is monotonic toward $s_0 \to 0$ under the $(1+z)^2$ scaling, so the reported value reflects the boundary, not a posterior peak.

†For Λcos, the likelihood maximum lies at the numerical prior floor $s_0 = 0.001$; the posterior median is 0.072, as reported in Sec. V B.

Acceptance fractions: A 0.65, B 0.62, C 0.65; all chains converged with $\tau_{\rm max} < 50$ for all parameters.

Each integer alternative fails for a distinct reason. Model A's $(1+z)^1$ scaling is too soft to reproduce matter dilution; the supernova sector tolerates this within $\Delta\chi^2_{\rm SN} \approx 55$, but the BAO sector adds a 164 penalty as the high-redshift bins (z = 1.32, 1.48, 2.33) reject the soft scaling. Model B saturates against the s_0 floor because the $(1+z)^2$ scaling offers no improvement over ΛCDM at any positive s_0; the BAO sector dominates the rejection at $\Delta\chi^2_{\rm BAO} \approx 1747$. Model C's $(1+z)^0$ scaling at high redshift is the most dramatic failure: with no decay of the matter-like term, both sectors reject the model ($\Delta\chi^2_{\rm SN} \approx 980$, $\Delta\chi^2_{\rm BAO} \approx 6300$).

Within the power-law clock family, the exponent n = -1/2 is selected analytically by three-dimensional matter dilution ($\rho_{\rm m} \propto S^{-3}$) through the Friedmann square root, giving the non-vacuum scaling $\mathcal{K}^{1/2} \propto S^{-3/2}$ at high z and the exact two-term non-vacuum kernel of Sec. II C at all z. The empirical fits confirm the analytic selection: among the alternatives tested, only n = -1/2 is viable.

TABLE X. Clock-exponent comparison from Appendix A: integer alternatives versus the matter-era selection $n = -1/2$.

Model	n	High-z scaling	Best-fit s_0	$H_0 r_d$ (km/s)	$\chi^2_{\rm SN}$	$\chi^2_{\rm BAO}$	$\chi^2_{\rm total}$	$\Delta\chi^2$ vs ΛCDM
A (proper time)	0	$(1+z)^1$	0.823	9279	1814.7	176.4	1991.1	+218.7
B (conformal)	-1	$(1+z)^2$	0.001*	8555	1834.5	1759.9	3594.4	+1821.9
C (symmetric)	+1	$(1+z)^0$	0.962	9220	2737.1	6312.0	9049.1	+7276.6
D (Λcos)	**-1/2**	$\mathbf{(1+z)^{3}/^{2}}$	$\mathbf{0.001^{\dagger}}$	**10010**	**1759.0**	**13.5**	**1772.6**	**+0.11**
ΛCDM (baseline)	—	—	(Ω_m = 0.312)	10044	1759.9	12.6	1772.5	0

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